EASY · ALGEBRA

Algebra

20

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ALGEBRA

John paid a total of $\$ 165$ for a microscope by making a down payment of $\$ 37$ plus p monthly payments of $\$ 16$ each. Which of the following equations represents this situation?

A. $16p - 37 = 165$

B. $37p - 16 = 165$

C. $16p + 37 = 165$

D. $37p + 16 = 165$

Q2

ALGEBRA

$$f(x) = x + b$$

For the linear function f, b is a constant. When $x = 0$, $f(x) = 30$. What is the value of b?

A. -30

B. $-\frac{1}{30}$

C. $\frac{1}{30}$

D. 30

Q3 ALGEBRA

Jay walks at a speed of 3 miles per hour and runs at a speed of 5 miles per hour. He walks for w hours and runs for r hours for a combined total of 14 miles. Which equation represents this situation?

A. $3w + 5r = 14$

B. $\frac{1}{3}w + \frac{1}{5}r = 14$

C. $\frac{1}{3}w + \frac{1}{5}r = 112$

D. $3w + 5r = 112$

Q4 ALGEBRA

If $x = 7$, what is the value of $x + 20$?

A. 13

B. 20

C. 27

D. 34

Q5

ALGEBRA

The function f is defined by $f x = \frac{1}{2}x + 6$. What is the value of $f 4$?

A. 20

B. 12

C. 10

D. 5

Q6 ALGEBRA

A line in the xy-plane has a slope of $-\frac{1}{2}$ and passes through the point $0, 3$. Which equation represents this line?

A. $y = -\frac{1}{2}x - 3$

B. $y = -\frac{1}{2}x + 3$

C. $y = \frac{1}{2}x - 3$

D. $y = \frac{1}{2}x + 3$

Q7 ALGEBRA

A gym charges its members a onetime $\$ 36$ enrollment fee and a membership fee of $\$ 19$ per month. If there are no charges other than the enrollment fee and the membership fee, after how many months will a member have been charged a total of $\$ 188$ at the gym?

A. 4

B. 5

C. 8

D. 10

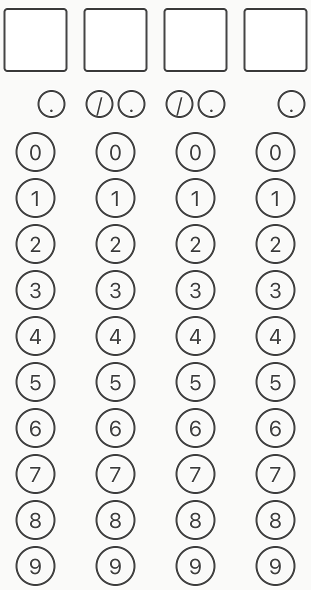
Q8

GRID-IN ALGEBRA

$$f(x) = 14 + 4x$$

The function f represents the total cost, in dollars, of attending an arcade when x games are played. How many games can be played for a total cost of $\$58$?

STUDENT-PRODUCED RESPONSE



Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9 ALGEBRA

A city's total expense budget for one year was x million dollars. The city budgeted y million dollars for departmental expenses and 201 million dollars for all other expenses. Which of the following represents the relationship between x and y in this context?

A. $x + y = 201$

B. $x - y = 201$

C. $2x - y = 201$

D. $y - x = 201$

Q10 ALGEBRA

3 more than 8 times a number x is equal to 83. Which equation represents this situation?

A. $(3)(8)x = 83$

B. $8x = 83 + 3$

C. $3x + 8 = 83$

D. $8x + 3 = 83$